Fundamentals Chapter 10

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1. Eigenvalues of T satisfy $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\mathbf{v}$. Let $\mathbf{v} = (v_1, v_2)^T$. We have that

$$T(\mathbf{v}) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{v} = \begin{bmatrix} v_2 \\ -v_1 \end{bmatrix} = \lambda\mathbf{v} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix}.$$

Solving the system gives us $v_2 = -\lambda^2 v_2$, so $1 + \lambda^2 = 0$. Since the field is $\mathbb{F}_5$, we have that $1 + \lambda^2 \equiv 0 \pmod 5$, so $\lambda = 2$ or $\lambda = 3$.

2. By the same process as problem 1, we have that $1 + \lambda^2 \equiv 0 \pmod p$. So any $\lambda \in \mathbb{F}_p$ satisfying $\lambda^2 \equiv -1 \pmod p$ is an eigenvalue.

3. Assume $\lambda = 0$. Then $A\mathbf{v} = \mathbf{0}$. Since A is invertible, we have that $\mathbf{v} = A^{-1}\mathbf{0} = \mathbf{0}$, but eigenvectors are nonzero. This is a contradiction, so $\lambda \neq 0$.

4. We have that $A\mathbf{v} = \lambda\mathbf{v}$, and since A is invertible, $\mathbf{v} = A^{-1}\lambda\mathbf{v}$, or $\lambda^{-1}\mathbf{v} = A^{-1}\mathbf{v}$, so λ^{-1} is an eigenvalue of A^{-1}.

5. By proposition 10.6, we know that the statement holds if $m = 1$. Now, assume it holds for k. That means, if

$$p(x) = (x - r_1) \cdots (x - r_k)q(x)$$

then

$$p(A) = (A - r_1 I_n) \cdots (A - r_k I_n)q(A).$$

By proposition 10.6, if $q(x) = (x - r_{k+1})q_1(x)$ then $q(A) = (A - r_{k+1}I_n)q_1(A)$. So

$$p(A) = (A - r_1 I_n) \cdots (A - r_k I_n)(A - r_{k+1}I_n)q_1(A).$$

As long as $q(x)$ factors, which in this case is until $q(x)$ is constant, $q(A)$ can be factored as well. By induction, if $p(x) = (x - r_1) \cdots (x - r_m)$ then $p(A) = (A - r_1) \cdots (A - r_m)$.

6. Let

$$T = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad S = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

We have that $ST = 0 = TS$. It seems that matrices that satisfy this property always have an eigenvalue of 0, and sometimes 1.

7. We already know that λ is an eigenvalue of T if and only if $T - \lambda 1$ is not injective.

By rank-nullity, if $T - \lambda 1$ is not injective, then $\dim(\text{null}(T - \lambda 1)) > 0$, so $\dim(\text{rank}(T - \lambda 1)) < \dim(V)$ (where V is the vector space), so $T - \lambda 1$ is not surjective. Similarly, $T - \lambda 1$ is not surjective implies that $\dim(\text{null}(T - \lambda 1)) > 0$, and $T - \lambda 1$ is not injective.

If $T - \lambda 1$ is not invertible, it's not injective, so λ is an eigenvalue of T. Finally, if λ is an eigenvalue of T, then $(T - \lambda 1)\mathbf{v} = \mathbf{0}$ for some nonzero $\mathbf{v}$. If $T - \lambda 1$ was invertible, then $\mathbf{v} = (T - \lambda 1)^{-1} \cdot \mathbf{0} = \mathbf{0}$, which is a contradiction. So $T - \lambda 1$ is not invertible.

9a.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}'.$$

9b. Consider the eigenvalues of A, λ_1, λ_2, and assume they are distinct. Let the corresponding eigenvectors be $\mathbf{v}_1, \mathbf{v}_2$. We can define a new variable $\mathbf{u}$ by expressing $\mathbf{x}$ in the basis of eigenvectors. Let $\mathbf{x} = P\mathbf{u}$ where

$$P = \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix}.$$

Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are distinct eigenvectors, they are not multiples of each other, so P is invertible. If we substitute $\mathbf{x} = P\mathbf{u}$ into $\mathbf{x}' = A\mathbf{x}$, we have

$$P\mathbf{u}' = AP\mathbf{u}, \text{ or } \mathbf{u} = P^{-1}AP\mathbf{u}.$$

By construction, $P^{-1}AP = D$ where D is a diagonal matrix of eigenvalues:

$$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix},$$

so the system becomes $\mathbf{u}' = D\mathbf{u}$. If we let $\mathbf{u} = \begin{bmatrix} w & z \end{bmatrix}^T$, we get the system

$$\begin{bmatrix} w' \\ z' \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} w \\ z \end{bmatrix},$$

which can be split into the two ODEs:

$$w' = \lambda_1 w, \quad z' = \lambda_2 z.$$

These can be solved independently and then reverted to the original basis to solve the original equation.